Ethan

Good afternoon, everyone. We're Group 23. Our project asks a simple question: can a reinforcement-learning controller make an office's air-conditioning greener? Not just by using less electricity, but by using it at the right time. Over the next fifteen minutes we'll show you the simulator we built, how we framed the problem, what our agent learned, and, just as importantly, what it didn't.

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Buildings use about thirty percent of the world's final energy, and in Singapore, air-conditioning is the biggest electricity user in most commercial buildings.

But not every kilowatt-hour is equally dirty. The carbon intensity of the grid changes hour to hour. When it's windy or sunny, more of the power is clean; in the evening peak, more of it comes from fossil fuels. A building can take advantage of that, because its walls and floors store coolness. You can cool a little earlier, while the grid is clean, and coast through the dirty hours.

That makes this a sequential decision problem: what you do at six in the morning affects how comfortable people are at two in the afternoon. So our question, on the slide, is whether an RL controller can beat a well-tuned rule, not just a basic thermostat, without making people uncomfortable.

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We built five pieces. A simulator of one office zone, driven by a year of weather, occupancy, price and carbon data. A formal decision problem, with a reward that combines carbon, cost and comfort. Two learning agents, PPO and DQN, tested against tuned rule-based controllers and an exact upper bound. A careful evaluation on months the agents never saw. And a responsible-AI analysis of what the agent learned, and who pays for it. Let's start with the simulator.

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We model one office zone as a 1R1C circuit. R is a thermal resistance: how well the walls keep the heat out. C is a capacitance: how much heat the building can soak up. Together they give a time constant of six hours. Switch the air-con off, and the room closes about two thirds of the gap to the outdoor temperature in six hours. Keep that number in mind, because it shapes every result you'll see.

The air-con has five power levels, up to fourteen kilowatts, and it gets less efficient on hot afternoons. We update the temperature exactly every fifteen minutes, and ten unit tests check the physics, the accounting and the energy scale.

Weather and office hours are Singapore-like. For carbon intensity we used a UK-like profile on purpose: Singapore's grid runs on about ninety-five percent natural gas, so its carbon intensity is almost flat, and there would be nothing to shift. All the data is synthetic. We train on January to September and test on 92 unseen days from October to December. I'll pass to my colleague, who'll explain how we turned this into a decision problem.

[next slide]

Brandon

Thanks. Here's the decision problem. Every fifteen minutes, the agent looks at twenty numbers: the indoor and outdoor temperature, an occupancy sensor, the electricity price, the carbon intensity, the time of day, whether it's a workday, and forecasts of temperature, price and carbon up to eight hours ahead. It then picks one of five power levels for the air-con. We kept the actions discrete on purpose: it means PPO, DQN and an exact planner all solve exactly the same problem, so the comparison is fair.

The reward is a penalty the agent tries to keep small. It adds up the carbon emitted, half the electricity cost, and a comfort penalty for every degree-hour outside 23 to 26 degrees, but only while people are actually in the office. The discount factor is 0.99, so at fifteen-minute steps the agent looks about a day ahead, long enough to link what it does now to comfort many hours later.

One observation shapes the rest of the talk. Carbon and cost both grow with energy, so a controller can cut carbon in two ways: use less energy, or use it at cleaner times. Only the second is truly carbon-aware. And in our data, price and carbon are almost unrelated, so the cheapest hour is not the cleanest one.

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Next, the controllers we compared. A fixed thermostat at 24 degrees. A setback that only cools during working hours. A carbon heuristic that cools deeper when the grid is clean and lets the room drift warmer when it's dirty. And a carbon-proportional rule, whose setpoint rises with carbon intensity.

The key word on this slide is tuned. Every rule was tuned on training days, 972 settings in total, using exactly the reward PPO maximises. That matters: our first results compared PPO with a carbon heuristic that was only partly tuned, and PPO looked eleven percent better. Once the rules were tuned fairly, most of that gap disappeared. In fact, all three rules ended up at the same answer: cool only when people are in, and hold 25.9 degrees, right at the top of the comfort band.

PPO got the same treatment. We tried twelve hyperparameter settings with two seeds each, picked the best on training days, and retrained it with five fresh seeds. Finally, the oracle. Because the physics is exact, we can solve each day perfectly by working backwards through it, if we know the future. No real controller can, so it's our upper bound. Everything is then tested once, on the 92 unseen days. I'll hand over to my colleague for the results.

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Jaylon

Thanks. Here's the main table. A fixed thermostat emits about nineteen kilograms of carbon a day. A simple setback at 24 degrees brings that down to eleven. Then just tuning that one setpoint, to 25.9 degrees, cuts another twenty-three percent, to 8.5 kilograms. That's the biggest single gain in the whole project, and it needs no learning at all.

Now the learning agents. PPO reaches 8.21 kilograms: 3.7 percent less carbon than the best tuned rule, with a tight confidence interval. It's also about one percent better on the reward it optimises, and that difference is statistically significant. Cost falls by about four percent. DQN doesn't beat the rule. The oracle, at 7.83, shows how much was possible, and PPO closes almost half of the carbon gap between the rule and the oracle.

But when we looked day by day, something stood out. Seventy-two percent of PPO's saving comes from just three public holidays. On those days the rule follows its workday schedule and cools an empty office, while PPO sees that nobody has arrived and stays off. On the other 89 days, PPO is only about one percent lower on carbon, and it's slightly worse on reward, because it pays for that one percent in comfort. Its comfort violations rise from under one percent to about five percent of occupied time, although each one is small, a fraction of a degree. A rule with a holiday calendar would capture the holiday saving without any learning.

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This is how the two agents learned. Each line is the average of five seeds, and the shaded band shows the best and worst seed. The dashed lines are the tuned rule and the oracle on the same training days.

PPO, in blue, learns smoothly. Early on, it learns to keep people comfortable, and then it spends the rest of training trimming energy and timing. By about three hundred thousand steps it has made ninety percent of its progress, and its five seeds end within 1.3 percent of each other on test carbon. DQN, in orange, learns about as quickly at first, but it collapses several times later in training, and its seeds disagree much more. That's why PPO is our main method. Tuning also changed how PPO learns: the default settings improve faster at first but settle lower, while the longer rollouts we picked trade some early speed for a better final policy.

Notice also where PPO ends: level with the tuned rule. That fits what we just saw. On ordinary days, there is very little between them. My colleague will now show you what PPO actually learned.

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Sean

Thanks. This is one test day, a Friday. The top panel is the indoor temperature, the middle one is the air-con power, and the bottom one is the grid's carbon intensity. The shaded area is when people are in.

Both controllers hold the room near the top of the comfort band, around 25.7 to 25.9 degrees, the same insight the tuning found. Neither pre-cools overnight: with a six-hour time constant, that coolness would leak away before anyone arrives. The difference is in the details. PPO, in blue, cools a little harder during the clean-grid dips around ten in the morning and over midday, then coasts up to 26 degrees as carbon rises towards the evening. So there is some genuine load-shifting. Across the test period, PPO pays 0.162 kilograms of carbon per kilowatt-hour, against 0.164 for the rule and 0.159 for the oracle. Real, but small.

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So where is the remaining room for improvement? We tested this by taking away one piece of the oracle's foresight at a time. If the oracle doesn't know future carbon intensity, it loses only 0.08 reward per day. If it doesn't know exactly when people arrive and leave, it loses about one full point, more than ten times as much, and it falls below even the tuned thermostat.

So in this building, the bottleneck isn't the carbon signal. It's knowing who is in the room. Everything else agrees: tuning switched off every carbon-reactive feature in the rules, removing carbon intensity from PPO's inputs changes its carbon by only about one percent, and PPO's own biggest win came from reacting to occupancy on holidays. With a six-hour time constant, one room is a leaky battery.

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Finally, the trade-off between carbon and comfort. This chart covers the 89 ordinary days, so neither side gets credit for the holidays. Each blue point is PPO trained with a different weight on comfort, and the green line is the thermostat with its setpoint swept from 24 to 26 degrees. Lower and further left is better.

At strict comfort, the tuned rule beats every PPO setting. PPO only wins in a narrow band, around a tenth of a degree-hour of discomfort per day, which the thermostat can't reach: at 25.9 degrees it's fine, but at 26 it's outside the band about a third of the time. If we relax comfort, PPO does learn real load-shifting, moving up to seventy-eight percent of its energy into the cleaner half of the day, but the discomfort that takes would not be acceptable in any real office. So where to sit on this curve is a value judgement, and my colleague will pick that up.

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Patrick

Thanks. Some things the numbers don't show on their own. First, who sets the weights? Our reward treats one degree-hour of discomfort as worth five kilograms of carbon, and nothing in the physics sets that rate. A facility manager would choose it, but the occupants live with it. That's why we show the whole trade-off curve, not one tuned point.

Second, specification gaming. PPO sits right at the edge of what the reward allows, and comfort only counts when the room is occupied, so a faulty sensor could be exploited. That's why we always score comfort on true occupancy, not on what the sensor says.

Third, comfort is a distribution, not an average. More than half of PPO's violations happen between five and half past six in the evening, so the people who stay late bear the cost.

Fourth, honest baselines. Our own first result overstated the gain about threefold, and a holiday calendar shrinks what's left to under one percent. Weak baselines are

a quiet way to mislead, so we report the fair comparison and where the gain comes from. Finally, transparency. A tuned thermostat gets most of the benefit, and anyone can explain it in one sentence. Choosing it over RL can be the responsible decision. Where RL is used, a person should stay in charge, with an override and a fallback to the rule.

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We also want to be clear about what this study can't claim yet. It's one zone with a simplified thermal model. The data is synthetic, and the carbon profile is UK-shaped, so these are not Singapore savings. PPO's margin is small, and DQN, the comfort sweep and the ablations used default PPO settings.

For the report, we're looking at occupancy prediction, since that's where the headroom is; robustness tests, such as shifting the carbon signal to check that the agent reads it rather than the clock; constrained PPO, which treats comfort as a hard limit instead of a weight; and a battery, to give the agent a real lever for shifting.

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To wrap up, four takeaways. First, tune your baseline: one setpoint gives twenty-three percent, and skipping that step inflated our RL result about threefold. Second, tuned PPO does win, by 3.7 percent on carbon, but mostly by staying off on holidays; on ordinary days it's about one percent, paid for in comfort. Third, in one room the carbon signal is worth little; knowing who is in the room is where the real headroom is. And fourth, it took an exact upper bound and fair tuning to see all of this. Thank you for listening. We're happy to take your questions.

Q&A (not read aloud)

Question	Owner	Answer
Is a 3.7% gain worth RL?	Sean / Patrick	Honestly, not in one room: 72% of it comes from 3 holidays, and against a rule with a holiday calendar PPO is −0.9% carbon and −1.8% reward. The value of the study is showing where the headroom is (occupancy) and when RL would pay (storage, multi-zone).
Isn't the holiday win legitimate?	Sean	Yes: PPO reacts to the occupancy sensor, which is useful. But any building system has a holiday calendar, so it isn't evidence of carbon-aware control. We report both views.
How did you make the comparison fair?	**Jaylon / Brandon**	Same training days, same reward, tuned budgets for both (972 rule settings; 12 PPO settings × 2 seeds), a single final test on held-out days, paired CIs over 5 seeds.
Why not continuous actions / SAC?	**Brandon**	Discrete actions let one problem admit PPO, DQN and an exact DP optimum. Five levels at 15-min steps already give 17 effective power levels per hour.
Why PPO over DQN?	**Brandon**	PPO is stable across seeds (±1.3%); DQN collapses intermittently and ends 7% worse on reward. DQN wasn't hyperparameter-tuned, which we state as a limitation.
How is the oracle optimal?	**Brandon**	Given the realised weather and occupancy, the dynamics are deterministic. Backward induction on a 0.05 °C grid with linear interpolation of V; the only approximation is the grid. Unit-tested to dominate every baseline.
What did tuning change in PPO?	**Brandon**	The rollout length doubled (n_steps 384 → 768 per env), giving larger, less noisy batches. γ = 0.995, smaller learning rates and bigger networks did not help.
Why γ = 0.99?	**Brandon**	1/(1−γ) = 100 steps ≈ 25 h, enough to link cooling now to comfort hours later. 0.995 was tested in tuning and was worse.
Isn't 5% violation bad?	Sean	It's frequent but tiny (0.08 °C·h/day, max ~0.14 °C) and mostly late in the day (53% between 17:00 and 18:30). At matched comfort the tuned thermostat wins; constrained PPO is the principled fix.
Why not real Singapore data?	Ethan	SG CI is nearly flat (gas grid), so there's nothing to shift. Loaders for real CSVs exist; synthetic profiles are calibrated to published ranges.
Is the agent just following the clock?	Sean	Mostly. Removing time features hurts most (reward −9%). But it pays 0.162 vs 0.164 kg/kWh, and without CI in its state that advantage mostly disappears, so it reacts to CI a little. The shifted-CI test in the report checks this directly.
Average vs marginal emissions?	Patrick	We use average intensity. Marginal factors could change which hours are "clean"; flagged as a limitation.